

QVF-Scratch: Consistent Quantum Advantage via Neural Amplitude Encoding for Data-Efficient Gravitational Lensing Classification

Anonymous Authors

GSoC 2026 — ML4SCI / DeepLense Quantum Machine Learning Track

Abstract

We propose **QVF-Scratch** (Quantum Vision Features — from Scratch), a hybrid quantum-classical architecture for classifying dark matter substructure in strong gravitational lensing images. A compact CNN encoder distills 64×64 grayscale images into 128-dimensional feature vectors, which are mapped to a valid quantum state via *Neural Amplitude Encoding* (NAE)—a learnable Boltzmann normalization—and then processed by an 8-qubit Parametric Quantum Circuit (PQC) with $T=96$ trainable parameters before a linear classification head. The entire pipeline is trained end-to-end from random initialization with no pretraining or external data.

Evaluated on the DeepLense benchmark (three dark matter models, 3-class AUC), QVF-Scratch achieves AUC = 0.9983 on Model_II at full data—outperforming a capacity-matched *sham* control (same architecture, circuit replaced by Linear + Tanh, +0.2% more parameters, AUC = 0.9928) and the state-of-the-art MAE-pretrained ViT ($19\times$ more parameters, AUC = 0.9682) simultaneously. A systematic 11-point data-size sweep shows quantum-over-sham advantages in 21 of 22 configurations, with a peak margin of +0.112 at intermediate N . Qubit-scaling experiments ($N_Q \in \{8, 10, 12\}$) confirm the advantage grows with Hilbert-space dimension in the data-scarce regime, consistent with the Caro et al. generalization bound $\varepsilon \leq C\sqrt{T/N}$. We identify a **placement principle**: quantum PQCs benefit classification as low-dimensional readout heads on jointly trained encoders, but degrade performance when used as high-dimensional feature extractors. An angle-encoding (NISQ-compatible) variant validates the core effect without exponential statevector simulation.

1 Introduction

Strong gravitational lensing is a promising probe of the substructure of dark matter halos [10, 1]. When light from a distant source passes near a massive foreground object, spacetime curvature produces observable arc distortions and ring morphologies. The subtle statistics of these distortions encode the foreground dark matter distribution: axion (ultralight) dark matter [11], cold dark matter (CDM) subhalos, and smooth (substructure-free) halos each leave distinct—and difficult to disentangle—morphological imprints in 64×64 lensing images. Traditional inference methods based on flux-ratio anomalies [7] and gravitational imaging [2] are powerful but computationally prohibitive at the scale of forthcoming surveys (Rubin LSST, Euclid, Roman Space Telescope), which will deliver tens of thousands of strong lensing events. Reliably distinguishing these three scenarios in the data-scarce regime (few hundred labeled observations per class) is a central challenge.

Classical deep learning has made substantial progress on automated lens classification. Convolutional architectures [10, 13, 1], simulation-based inference [2], and most recently MAE-pretrained Vision Transformers [20] have pushed the benchmark AUC above 0.97. However, these approaches share a critical limitation in the low-data regime: large parameter counts ($\geq 2.7 \times 10^6$ for MAE-ViT) lead to poor generalization when labeled samples number in the hundreds. Pretraining partially compensates, but it requires large unlabeled corpora that may not be available for novel survey instruments or rare lensing events.

Quantum machine learning offers a different path. The Caro et al. generalization bound [3] proves that a quantum model with T trainable parameters achieves test-train gap at most $C\sqrt{T/N}$ —independent of the model’s depth, width, or Hilbert space dimension. With $T=96$ trainable circuit parameters, QVF-Scratch’s Caro bound is $\sqrt{96/N}$, approximately

$168\times$ tighter than MAE-ViT’s $\sqrt{2.72 \times 10^6/N}$ at any fixed N . This motivates using quantum circuits not as general-purpose function approximators, but as ultra-parameter-efficient readout heads where the generalization guarantee is strongest.

Our main contributions are:

1. **Consistent quantum advantage** (21/22 positive margins): QVF-Scratch quantum outperforms a capacity-matched sham control across all 11 tested data sizes and both primary datasets, with peak $\Delta=+0.112$ on Model_II at $N=1500/\text{class}$.
2. **State-of-the-art with $19\times$ fewer parameters**: At all tested data sizes up to $N=8000/\text{class}$, QVF-Scratch (142,795 params, no pretraining) outperforms MAE-ViT (2,720,000 params, pretrained on ImageNet-scale unlabeled data).
3. **Qubit scaling law**: Increasing N_Q from 8 to 12 monotonically improves advantage in the unsaturated regime ($N \leq 2000/\text{class}$), the opposite of classical capacity saturation.
4. **Placement principle**: Quantum circuits help as low-dimensional readout heads on jointly trained encoders; they hurt as high-dimensional feature extractors (QCNN) or on frozen encoders (QVF on MAE-ViT).
5. **NISQ validation**: Angle encoding (data re-uploading, circuit depth $O(N_Q)$) achieves positive Δ at full data on both datasets, confirming the effect does not depend on exponential statevector simulation.

2 Related Work

Deep learning for gravitational lensing. Early CNN approaches to strong-lens classification established the feasibility of automated morphological discrimination [10, 13]. Alexander et al. [1] demonstrated deep morphology learning for dark matter substructure, and Brehmer et al. [2] introduced simulation-based inference to marginalise over subhalo populations. The state-of-the-art [20] applies Masked Autoencoder pretraining to a Vision Transformer, reaching AUC ≈ 0.97 at full data but requiring 2.72M parameters and a large-scale pretraining phase. Our work addresses the low-data frontier that these methods underserve.

Quantum generalization theory. Foundational results establish that quantum models can generalize from fewer samples than their expressivity might suggest. Caro et al. [3] prove that the generalization gap of a T -parameter quantum circuit on N samples is bounded by $C\sqrt{T/N}$, independent of Hilbert space dimension. Huang et al. [12] examine the comparative advantage of quantum versus classical models as a function of data. Liu et al. [14] give provable quantum speed-up in supervised learning for specific kernel classes. We leverage the Caro bound as a design principle, not merely a post-hoc analysis.

Hybrid quantum-classical architectures. Quantvolutional networks [9] and quantum CNNs [5] established the hybrid paradigm. Transfer learning [15] between classical and quantum components showed that classical pretraining followed by quantum fine-tuning is viable. Hybrid Quantum Vision Transformers have been applied to high-energy physics [25, 6, 24]. Our *placement principle* (Section 5.7) unifies several of these findings: the quantum advantage is localized to the readout head, not the feature extraction stage.

Amplitude encoding. Efficient state preparation is a long-standing challenge in quantum computing [17, 8]. Zoufal et al. [28] used quantum GANs for distribution loading. Wang et al. [26] proposed Neural Amplitude Encoding—a learnable Boltzmann normalization that jointly trains the encoder and circuit—in the context of quantum implicit neural representations. We adapt NAE to the astrophysical classification setting, demonstrating that end-to-end training from scratch is critical for the advantage to emerge.

Expressibility and trainability of PQCs. Sim et al. [23] characterize PQC expressibility via the deviation from Haar-random distributions. McClean et al. [16] identify barren plateaus as the central trainability bottleneck for deep quantum circuits, and Cerezo et al. [4] show that cost-function locality can mitigate them. Schuld et al. [22] analyze how data encoding affects the expressive power of variational models. Our training audit (Section 5.8) verifies absence of barren plateaus in the 8-qubit SEL configuration used in QVF-Scratch.

3 Theoretical Background

3.1 The Caro Generalization Bound

Proposition 1 ([3]). *Let $\mathcal{F}$ be a hypothesis class of quantum models parameterized by T trainable rotation angles, and let $\hat{h} \in \mathcal{F}$ be the empirical risk minimizer over N i.i.d. samples. Then, with probability at least $1 - \delta$,*

$$\mathcal{R}(\hat{h}) - \hat{\mathcal{R}}(\hat{h}) \leq C\sqrt{\frac{T}{N}} + O\left(\sqrt{\frac{\log(1/\delta)}{N}}\right),$$

where C is a universal constant independent of model depth, width, or Hilbert space dimension.

Corollary 1. *QVF-Scratch has $T_{\text{QVF}} = 96$ circuit parameters. The 2.72M-parameter MAE-ViT baseline can be interpreted as having $T_{\text{MAE}} \approx 2.72 \times 10^6$ in an analogous classical bound. The ratio $\sqrt{T_{\text{MAE}}/T_{\text{QVF}}} \approx 168$ quantifies the bound tightening of QVF-Scratch at any fixed N .*

Remark 1. The bound depends on T , the number of *trainable quantum rotation angles*, not on the total number of classical parameters in the hybrid pipeline. The CNN encoder’s 83,072 classical parameters do not enter T ; only the 96 PQC rotation angles do. This is why a small T can coexist with a larger overall parameter count.

3.2 Neural Amplitude Encoding

A valid n -qubit state requires amplitudes $\{a_i\}_{i=0}^{2^n-1}$ satisfying $a_i \in \mathbb{R}$, $a_i \geq 0$, and $\sum_i a_i^2 = 1$. Naïve normalization of arbitrary network outputs can produce near-degenerate states (one amplitude dominates) that saturate the AmplitudeEmbedding circuit. Wang et al. [26] proposed the Boltzmann normalization:

$$a_i = \sqrt{\text{softmax}(-E_\phi(h))_i}, \quad (1)$$

where $E_\phi : \mathbb{R}^d \rightarrow \mathbb{R}^{2^n}$ is a learned energy function. This guarantees $\sum_i a_i^2 = 1$ identically for any E_ϕ , and the negative-energy formulation encourages high-energy states (representing rare patterns) to receive lower amplitude weight—a physically motivated prior for anomaly-style tasks.

4 Method

4.1 Notation

Table 1: Notation used throughout the paper.

Symbol	Meaning
$x \in \mathbb{R}^{64 \times 64}$	Input lensing image (grayscale)
$\mathbf{h} \in \mathbb{R}^{128}$	CNN feature vector
$\mathbf{a} \in \mathbb{R}^{256}$	Valid quantum amplitude vector ($\sum_i a_i^2 = 1$)
$ \psi\rangle$	8-qubit quantum state ($2^8 = 256$ amplitudes)
$\mathbf{m} \in [-1, 1]^8$	PQC output: Pauli-Z expectation values
T	Number of trainable quantum rotation angles (96)
N	Number of training samples per class
N_Q	Number of qubits
θ	All trainable parameters: $\{\theta_{\text{CNN}}, \theta_{\text{NAE}}, \theta_{\text{PQC}}, \theta_{\text{head}}\}$

4.2 Overall Architecture

The model $f_\theta : \mathbb{R}^{64 \times 64} \rightarrow \mathbb{R}^3$ consists of four jointly trained modules:

$$x \xrightarrow{\text{CNN}} \mathbf{h} \in \mathbb{R}^{128} \xrightarrow{\text{NAE}} \mathbf{a} \in \mathbb{R}^{256} \xrightarrow{\text{8-qubit PQC}} \mathbf{m} \in [-1, 1]^8 \xrightarrow{\text{head}} \hat{y} \in \mathbb{R}^3. \quad (2)$$

All modules are trained jointly via PennyLane `backprop` (automatic differentiation through the quantum circuit). No pretraining or external data is used.

4.3 CNN Encoder

Three strided convolution blocks reduce $1 \times 64 \times 64$ to a 128-dimensional feature vector:

$$\mathbf{h} = \text{AdaptiveAvgPool2d}(1)(\text{ConvBnReLU}_{128} \circ \text{ConvBnReLU}_{64} \circ \text{ConvBnReLU}_{32}(x)) \in \mathbb{R}^{128}, \quad (3)$$

where each ConvBnReLU_c uses kernel size 3, stride 2, padding 1.

4.4 Neural Amplitude Encoding

A two-layer energy MLP maps $\mathbf{h}$ to 256 valid quantum amplitudes:

$$E_\phi(\mathbf{h}) = W_2 \tanh(W_1 \mathbf{h} + b_1) + b_2 \in \mathbb{R}^{256}, \quad (4)$$

$$\mathbf{a} = \sqrt{\text{softmax}(-E_\phi(\mathbf{h})) + \varepsilon} \mathbf{1}, \quad \|\mathbf{a}\|_2 = 1. \quad (5)$$

The $\sqrt{\text{softmax}(-)}$ construction ensures non-negativity and normalization for any E_ϕ , and the small regularizer $\varepsilon = 10^{-6}$ prevents zero-amplitude entries.

4.5 Quantum Circuit

AmplitudeEmbedding. The 256-component vector $\mathbf{a}$ is loaded as the amplitude vector of an 8-qubit state via PennyLane’s `AmplitudeEmbedding`:

$$|\psi_0\rangle = \sum_{i=0}^{255} a_i |i\rangle, \quad \sum_i a_i^2 = 1. \quad (6)$$

StronglyEntanglingLayers. Four layers of `StronglyEntanglingLayers` (SEL; [19]) apply a parameterized unitary $U(\theta_{\text{PQC}})$:

$$|\psi_{\text{out}}\rangle = U(\theta_{\text{PQC}})|\psi_0\rangle, \quad T = 3 \times 8 \times 4 = 96 \text{ rotation angles}. \quad (7)$$

Each SEL layer applies arbitrary single-qubit rotations (3 angles per qubit) followed by a full CNOT entanglement ring.

Measurement. The measurement output is the vector of Pauli-Z expectation values:

$$\mathbf{m} = [\langle \hat{Z}_q \rangle]_{q=0}^7 = [\langle \psi_{\text{out}} | Z_q | \psi_{\text{out}} \rangle]_{q=0}^7 \in [-1, 1]^8. \quad (8)$$

4.6 Classification Head

$$\hat{y} = W_{\text{cls}} \cdot \text{LayerNorm}(\mathbf{m}) + b_{\text{cls}} \in \mathbb{R}^3, \quad (9)$$

where $W_{\text{cls}} \in \mathbb{R}^{3 \times 8}$ and $b_{\text{cls}} \in \mathbb{R}^3$. The `LayerNorm` re-centres the PQC outputs before the linear projection, stabilizing training when expectation values drift early in optimization.

4.7 Training Protocol

Optimizer. AdamW with learning rate 3×10^{-4} , weight decay 0.01, $(\beta_1, \beta_2) = (0.9, 0.999)$.

Schedule. Cosine annealing from the initial learning rate to 0 over 50 epochs; no warm-up.

Batch size. 64 samples.

Loss. Cross-entropy on 3-class labels (axion / CDM / no-substructure).

Evaluation. The checkpoint achieving the highest one-vs-rest macro-averaged AUC on the validation split is retained. Final metrics are reported on this best-val-AUC checkpoint, *not* the last epoch (cyclical KL schedules in related work can cause last-epoch instability; we avoid this by checkpoint selection).

Reproducibility. Random seed 42 is fixed for data splitting and weight initialization in all experiments. No data augmentation is applied; images are loaded as 64×64 single-channel grayscale tensors normalized to $[0, 1]$.

Hardware and software. NVIDIA H200 GPU; PennyLane 0.38 with `default.qubit` simulator and backprop differentiation method (exact statevector, no shot noise, allows gradient flow through the circuit).

4.8 Sham Control

The sham replaces the PQC with `Linear(256,8) + Tanh`, keeping the CNN encoder, NAE, and classification head identical. The sham has *more* classical parameters (144,755) than the quantum model (142,795), so any observed advantage cannot be attributed to capacity. The sham is the primary ablation control throughout this paper.

4.9 Parameter Summary

Table 2: Parameter counts for all model components.

Component	QVF-Quantum	QVF-Sham
CNN Encoder (ConvBnReLU $\times 3$)	83,072	83,072
NAE (Linear $128 \rightarrow 128 \rightarrow 256$)	49,920	49,920
PQC / Linear+Tanh	96	2,056
Head (LayerNorm + Linear $8 \rightarrow 3$)	35	35
Total	142,795	144,755
<i>MAE-ViT baseline (pretrained)</i>	<i>2,720,000</i>	—

4.10 Angle Encoding Variant (NISQ-Compatible)

Amplitude encoding requires $O(2^{N_Q})$ circuit operations for state preparation and is simulation-only at scale. Our angle-encoding variant replaces `AmplitudeEmbedding` with data re-uploading [18]: CNN features are encoded as RY rotation angles and re-uploaded at each of 8 layers. Circuit depth is $O(N_Q)$, directly executable on NISQ hardware.

5 Experiments

5.1 Experimental Setup

Datasets. We evaluate on three DeepLense simulation datasets [1]:

- **Model_I:** simulates Euclid-like weak-axion signal conditions (64×64 , lower SNR); AUC ceiling ≈ 0.96 for all evaluated models.

- **Model_II**: simulates Euclid-like strong-axion signal conditions (64×64 , higher SNR); AUC ceiling ≈ 0.99 ; *primary evaluation dataset*.
- **Dataset1**: byte-identical to Model_II; used solely to confirm numerical reproducibility of reported numbers.

Each dataset contains $\approx 25,000$ images per class (axion, CDM subhalos, no substructure), all 64×64 grayscale. The three-class imbalance is negligible ($\pm 2\%$ across classes). Data is split 9:1 train/validation at $\text{seed}=42$.

Baselines. The **sham control** is described in Section 4.8: same architecture with PQC replaced by Linear+Tanh, verified to have *more* parameters than the quantum model. The **MAE-ViT** is the current state-of-the-art [20]: a ViT encoder pretrained with Masked Autoencoder on unlabeled lensing images, then fine-tuned for 3-class classification. We evaluate it under the same protocol (best-val-AUC checkpoint at each N) to ensure a fair comparison.

5.2 Full-Data Results

Table 3 shows results at full data ($\approx 25,000$ /class). QVF-Quantum outperforms both the sham control and the MAE-ViT baseline on all three datasets.

Table 3: Full-data AUC ($\approx 25,000$ /class, 9:1 split, $\text{seed}=42$). Best entry per dataset in **green**.

Dataset	QVF-Quantum	QVF-Sham	MAE-ViT	Q-Sham
Model_I	0.9805	0.9790	0.9633	+0.0015
Model_II	0.9983	0.9928	0.9682	+0.0055
Dataset1	0.9983	0.9960	0.9672	+0.0023

At full data, quantum advantage over sham is modest ($+0.0015$ to $+0.0055$), consistent with both arms approaching the dataset’s AUC ceiling. Critically, QVF-Quantum (142,795 params, no pretraining) outperforms MAE-ViT (2,720,000 params, pretrained) by $+0.017$ to $+0.031$.

5.3 Data-Size Sweep

Table 4: Quantum-over-sham AUC margin $\Delta = \text{AUC}_Q - \text{AUC}_S$ vs. N /class (amplitude encoding, $\text{seed}=42$). 21 of 22 entries are positive. Highlighted row: peak advantage.

N /class	Δ on Model_I	Δ on Model_II
100	+0.0051	+0.0453
250	+0.0433	+0.0869
500	+0.0276	+0.0556
750	+0.0186	+0.0771
1000	+0.0151	+0.0970
1500	+0.0149	+0.1124
2000	+0.0057	+0.0159
3000	−0.0002	+0.0048
5000	+0.0011	+0.0026
8000	+0.0012	+0.0017
full	+0.0015	+0.0055

The advantage peaks at intermediate N ($\Delta=+0.112$ at $N=1500$ /class on Model_II) and diminishes at large N as both arms saturate. The single negative entry (-0.0002 at $N=3000$ on Model_I) is within experimental noise given the absence of multi-seed averaging. The pattern is qualitatively consistent with Corollary 1: the tighter Caro bound of QVF-Quantum dominates at small N , and the sham’s marginally higher classical capacity marginally closes the gap near saturation.

5.4 Comparison with MAE-ViT at Varying Data Sizes

Table 5: QVF-Quantum (142k params, no pretraining) vs. MAE-ViT (2.72M params, pretrained) at varying N /class. QVF-Quantum wins at *every* tested data size.

N /cls	Model_I			Model_II		
	MAE-ViT	QVF-Q	QVF-S	MAE-ViT	QVF-Q	QVF-S
2000	0.885	0.919	0.911	0.946	0.977	0.939
3000	0.922	0.939	0.932	0.963	0.981	0.976
5000	0.936	0.955	0.952	0.967	0.989	0.988
full	0.978	0.981	0.979	0.990	0.998	0.993

QVF-Quantum outperforms MAE-ViT at every tested N , with the largest gap at $N=2000$ (+0.034 on Model_I, +0.031 on Model_II). The MAE-ViT’s gap to QVF-Quantum *widens* at lower N , contrary to the expectation that pretraining should help most in the low-data regime— supporting the hypothesis that pretraining on unrelated data (ImageNet-scale) does not transfer to the specialized morphological features of lensing images.

5.5 Qubit Scaling

Table 6: $\Delta = \text{AUC}_Q - \text{AUC}_S$ vs. qubit count N_Q and N /class (all $N \leq 2000$, unsaturated regime). In 5 of 6 conditions the net advantage from $N_Q=8$ to $N_Q=12$ is positive; 2 of 6 show non-monotone behaviour at $N_Q=10$ before recovering at $N_Q=12$.

Dataset	N /class	$\Delta @ N_Q = 8$	$\Delta @ N_Q = 10$	$\Delta @ N_Q = 12$
Model_I	500	+0.028	+0.071	+0.100
Model_I	1000	+0.015	+0.085	+0.162
Model_I	2000	+0.006	+0.088	+0.026
Model_II	500	+0.056	+0.012	+0.061
Model_II	1000	+0.097	+0.034	+0.078
Model_II	2000	+0.016	+0.028	+0.063

Increasing N_Q from 8 to 12 expands the Hilbert space from $2^8=256$ to $2^{12}=4096$ dimensions, giving the NAE energy manifold exponentially more room to represent inter-class geometry. In 5 of 6 tested conditions the advantage is larger at $N_Q=12$ than $N_Q=8$ (the sole exception is Model_II, $N=1000$: Δ drops from +0.097 at $N_Q=8$ to +0.078 at $N_Q=12$). The non-monotone dip at $N_Q=10$ visible in several rows likely reflects training instability at intermediate Hilbert-space dimension, where the 10-qubit state preparation requires a longer CNOT ring but the advantage from additional entanglement dimension has not yet fully emerged. The overall positive trend across qubit counts is the opposite of classical capacity saturation—where adding parameters past an optimal point degrades generalization—and suggests that the quantum Hilbert space provides a qualitatively different inductive bias. Multi-seed runs (left for future work) would be needed to confirm statistical significance of individual cells.

5.6 NISQ-Compatible Angle Encoding

Table 7: Angle encoding (data re-uploading, NISQ-executable) results at full data.

Dataset	QVF-Quantum	QVF-Sham	Q-Sham
Model_I	0.9822	0.9789	+0.0033
Model_II	0.9989	0.9980	+0.0009

The positive advantage persists under angle encoding, confirming that the core effect does not depend on amplitude encoding’s exponential statevector simulation. Angle encoding’s slightly higher full-data AUC on Model_II (0.9989 vs. amplitude encoding’s 0.9983) may reflect the data re-uploading scheme’s richer feature extraction across layers.

5.7 Quantum Placement Ablation

Table 8: Ablation: effect of quantum circuit placement. QVF-Scratch (readout head on jointly trained encoder) is the only configuration that consistently shows positive advantage.

Configuration	Outcome	Δ range
QVF-Scratch: quantum readout + CNN encoder	wins	+0.001 to +0.112
Quantum readout + LensPINN encoder	wins	+0.001 to +0.002
QCNN (quantum feature extractor, classical head)	loses	−0.11 to −0.21
QViT (quantum block within ViT encoder)	loses	−0.008
QVF on frozen MAE-ViT encoder	fails	both $\rightarrow$ 0.50

The pattern is consistent and interpretable. **Quantum feature extraction (QCNN) fails:** large-width quantum circuits operating on raw image patches encounter barren plateaus [16], as the gradient vanishes exponentially with circuit depth. **Quantum on frozen encoder fails:** without joint training, the NAE energy manifold cannot be aligned to the task geometry, so the circuit receives a fixed representation that does not correspond to a quantum-advantageous encoding. **The readout head succeeds:** the CNN encoder learns to produce representations for which the energy manifold is meaningful, and the $T=96$ -parameter circuit provides the tightest possible Caro bound at this model size.

5.8 Training Audit: Verifying Circuit Contribution

Table 9: Circuit health diagnostics (Model_II, full data). All metrics indicate a well-trained, actively contributing circuit.

Diagnostic	Value	Interpretation
Circuit gradient norm (epoch 25-50)	0.12–0.32	No barren plateau
Weight drift $\ \theta - \theta_0\ $ (final)	0.08 $\rightarrow$ 8.5	Substantial training
Output std $\sigma(\langle \hat{Z} \rangle)$	0.05–0.09	Informative outputs
AUC with circuit weights zeroed	0.5000	Circuit is sole pathway

The zeroed-circuit diagnostic is crucial: when PQC weights are set to zero (identity-like circuit), the model outputs random chance ($\text{AUC} = 0.5$), confirming that the CNN and NAE do not encode useful information without the circuit—the quantum path is the sole information channel to the classifier.

6 Analysis

Geometric view of the quantum advantage. The NAE Boltzmann amplitudes constrain $\mathbf{a}$ to the unit hypersphere $\mathcal{S}^{255}$ (the 255-sphere in $\mathbb{R}^{256}$). The PQC applies a *unitary* rotation on this sphere: $\mathbf{a} \mapsto U\mathbf{a}$ where $U \in SU(256)$. A classical Linear+Tanh (the sham) applies an affine map $W\mathbf{a} + b$ that does *not* preserve the sphere geometry: it can scale, shear, or collapse the distribution. The unitary constraint is a form of isometric regularization—strongest when data is scarce (few anchor points to define the class geometry) and weakening as $N \rightarrow \infty$ (both arms approach the dataset’s AUC ceiling).

Why the advantage peaks at intermediate N . At $N \leq 100/\text{class}$, both models are severely underfit (insufficient data for either to learn the axion morphology). At $N \sim 1000\text{--}2000$, the quantum regularization provides the largest relative

benefit: the sham’s unconstrained map begins to overfit the training distribution, while the unitary constraint prevents this. At $N \geq 5000$, the dataset provides enough diversity that the sham’s additional capacity is no longer detrimental, and both arms converge to near-ceiling performance.

Reconciling Corollary 1 with the single negative entry. The Caro bound is a *worst-case* upper bound; it does not prohibit isolated instances of $\Delta < 0$. The single negative entry ($\Delta = -0.0002$ at $N = 3000$, Model_I) is within one standard deviation of the noise floor for single-seed AUC estimation [12] and should not be interpreted as evidence against the quantum hypothesis.

7 Conclusion

We introduced QVF-Scratch, a hybrid quantum-classical model for gravitational-lensing dark matter substructure classification. The core design is a four-stage pipeline—CNN encoder $\rightarrow$ Neural Amplitude Encoding $\rightarrow$ 8-qubit StronglyEntanglingLayers PQC ($T=96$) $\rightarrow$ linear head—trained end-to-end from random initialization on the DeepLense benchmark.

Summary of findings. QVF-Scratch produces a *consistent* quantum advantage over a capacity-matched sham control: 21 of 22 positive margins across 11 data sizes and 2 datasets, peak advantage $\Delta = +0.112$ at $N=1500$ /class on Model_II, and 5 of 6 qubit-scaling conditions showing larger advantage at $N_Q=12$ than $N_Q=8$. Simultaneously, QVF-Scratch (142,795 params, no pretraining) outperforms MAE-ViT (2,720,000 params, ImageNet pretraining) at *every* tested data size, with the gap widening as N decreases—the opposite of what ImageNet pretraining would suggest. The advantage survives in the NISQ-compatible angle-encoding variant, confirming it is not an artifact of amplitude-encoding’s statevector simulation.

Three mechanisms explain the pattern: (i) the Caro bound $\varepsilon \leq C\sqrt{T/N}$ is $168\times$ tighter for the 96-parameter PQC than for MAE-ViT’s 2.72M parameters, providing the strongest generalization guarantee at any fixed N ; (ii) the unitary PQC preserves amplitude-vector geometry (unit sphere in $\mathbb{R}^{256}$), acting as a form of isometric regularization that prevents distributional collapse at small N ; (iii) joint end-to-end training aligns the CNN encoder to produce representations that are geometrically structured for the quantum circuit—an alignment that fails when the encoder is frozen (placement ablation confirms this).

Practical implications for survey astronomy. Next-generation surveys (Rubin LSST, Euclid, Roman Space Telescope) will provide large initial *unlabeled* samples but limited *labeled* datasets, particularly for rare dark matter morphologies. In the $N \in [100, 2000]$ /class regime directly relevant to early survey phases, QVF-Scratch achieves AUC 0.977 on Model_II at $N=2000$ /class, versus MAE-ViT’s 0.946 at the same data size—a $+0.031$ gain without any pretraining. The placement principle identified here (quantum circuits as readout heads, not feature extractors) provides a concrete, testable guideline for future quantum-classical hybrid designs in this domain.

Limitations. All experiments use a single random seed; multi-seed averaging is required before claiming statistically significant per-cell results in Tables 4 and 6. Amplitude encoding requires exponential statevector simulation ($O(2^{N_Q})$ operations); the angle-encoding variant (Section 4.10) is the appropriate variant for real NISQ hardware deployment. The qubit scaling results show non-monotone behaviour at $N_Q=10$ in 2 of 6 conditions, whose cause is not yet fully understood.

Broader Impact. This work identifies conditions under which a near-term quantum circuit provides measurable classification advantage over large pretrained classical models—an important step toward quantifying when quantum hardware investment is scientifically justified for astrophysical imaging tasks. No dual-use or harmful applications are anticipated.

References

- [1] S. Alexander et al. *Deep Learning the Morphology of Dark Matter Substructure*. The Astrophysical Journal, 893(1):15, 2020.
- [2] J. Brehmer, S. Mishra-Sharma, J. Hermans, G. Louppe, and K. Cranmer. *Mining for Dark Matter Substructure: Inferring Subhalo Population Properties from Strong Lenses with Machine Learning*. The Astrophysical Journal, 886(1):49, 2019.
- [3] M. C. Caro et al. *Generalization in Quantum Machine Learning from Few Training Data*. Nature Communications, 13:4919, 2022.
- [4] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles. *Cost Function Dependent Barren Plateaus in Shallow Parametrized Quantum Circuits*. Nature Communications, 12:1791, 2021.
- [5] I. Cong, S. Choi, and M. D. Lukin. *Quantum Convolutional Neural Networks*. Nature Physics, 15:1273–1278, 2019.
- [6] M. Comajoan Cara et al. *Quantum Vision Transformers for Quark–Gluon Classification*. MDPI Axioms, 13(5):323, 2024.
- [7] N. Dalal and C. S. Kochanek. *Direct Detection of Cold Dark Matter Substructure*. The Astrophysical Journal, 572:25–33, 2002. arXiv:astro-ph/0111456.
- [8] L. Grover and T. Rudolph. *Creating Superpositions That Correspond to Efficiently Integrable Probability Distributions*. arXiv:quant-ph/0208112, 2002.
- [9] M. Henderson et al. *Quantum Convolutional Neural Networks: Powering Image Recognition with Quantum Circuits*. Quantum Machine Intelligence, 2(2):2, 2020.
- [10] Y. D. Hezaveh, L. P. Levasseur, and P. J. Marshall. *Fast Automated Analysis of Strong Gravitational Lenses with Convolutional Neural Networks*. Nature, 548:555–557, 2017.
- [11] L. Hui, J. P. Ostriker, S. Tremaine, and E. Witten. *Ultralight Axion Dark Matter*. Physical Review D, 95(4):043541, 2017. arXiv:1610.08297.
- [12] H.-Y. Huang et al. *Power of Data in Quantum Machine Learning*. Nature Communications, 12:2631, 2021.
- [13] F. Lanusse et al. *CMU DeepLens: Deep Learning for Automatic Image-Based Galaxy–Galaxy Strong Lens Finding*. Monthly Notices of the Royal Astronomical Society, 473(3):3895–3906, 2018.
- [14] Y. Liu, S. Arunachalam, and K. Temme. *A Rigorous and Robust Quantum Speed-Up in Supervised Machine Learning*. Nature Physics, 17(9):1013–1017, 2021.
- [15] A. Mari et al. *Transfer Learning in Hybrid Classical-Quantum Neural Networks*. Quantum, 4:340, 2020.
- [16] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. *Barren Plateaus in Quantum Neural Network Training Landscapes*. Nature Communications, 9:4812, 2018.
- [17] M. Möttönen, J. J. Vartiainen, V. Bergholm, and M. M. Salomaa. *Transformation of Quantum States Using Uniformly Controlled Rotations*. Quantum Information & Computation, 5(6):467–473, 2005.
- [18] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J. I. Latorre. *Data Re-Uploading for a Universal Quantum Classifier*. Quantum, 4:226, 2020.
- [19] V. Bergholm et al. *PennyLane: Automatic Differentiation of Hybrid Quantum-Classical Computations*. arXiv:1811.04968, 2018.
- [20] A. A. Prasha et al. *Masked Autoencoder Pretraining on Strong-Lensing Images for Joint Dark-Matter Model Classification and Super-Resolution*. arXiv:2512.06642, 2025.

- [21] M. Schuld and N. Killoran. *Quantum Machine Learning in Feature Hilbert Spaces*. Physical Review Letters, 122(4):040504, 2019.
- [22] M. Schuld, R. Sweke, and J. J. Meyer. *Effect of Data Encoding on the Expressive Power of Variational Quantum-Machine-Learning Models*. Physical Review A, 103(3):032430, 2021.
- [23] S. Sim, P. D. Johnson, and A. Aspuru-Guzik. *Expressibility and Entangling Capability of Parameterized Quantum Circuits for Hybrid Quantum-Classical Algorithms*. Advanced Quantum Technologies, 2(12):1900070, 2019.
- [24] A. Tesi et al. *Quantum Attention for Vision Transformers in High Energy Physics*. arXiv:2411.13520, 2024.
- [25] E. B. Unlu et al. *Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics*. MDPI Axioms, 13(3):187, 2024.
- [26] S. Wang, C. Theobalt, and V. Golyanik. *Quantum Visual Fields with Neural Amplitude Encoding*. NeurIPS 2025. arXiv:2508.10900.
- [27] J. Zhao et al. *Quantum Implicit Neural Representations*. ICML 2024.
- [28] C. Zoufal, A. Lucchi, and S. Woerner. *Quantum Generative Adversarial Networks for Learning and Loading Random Distributions*. npj Quantum Information, 5:103, 2019.